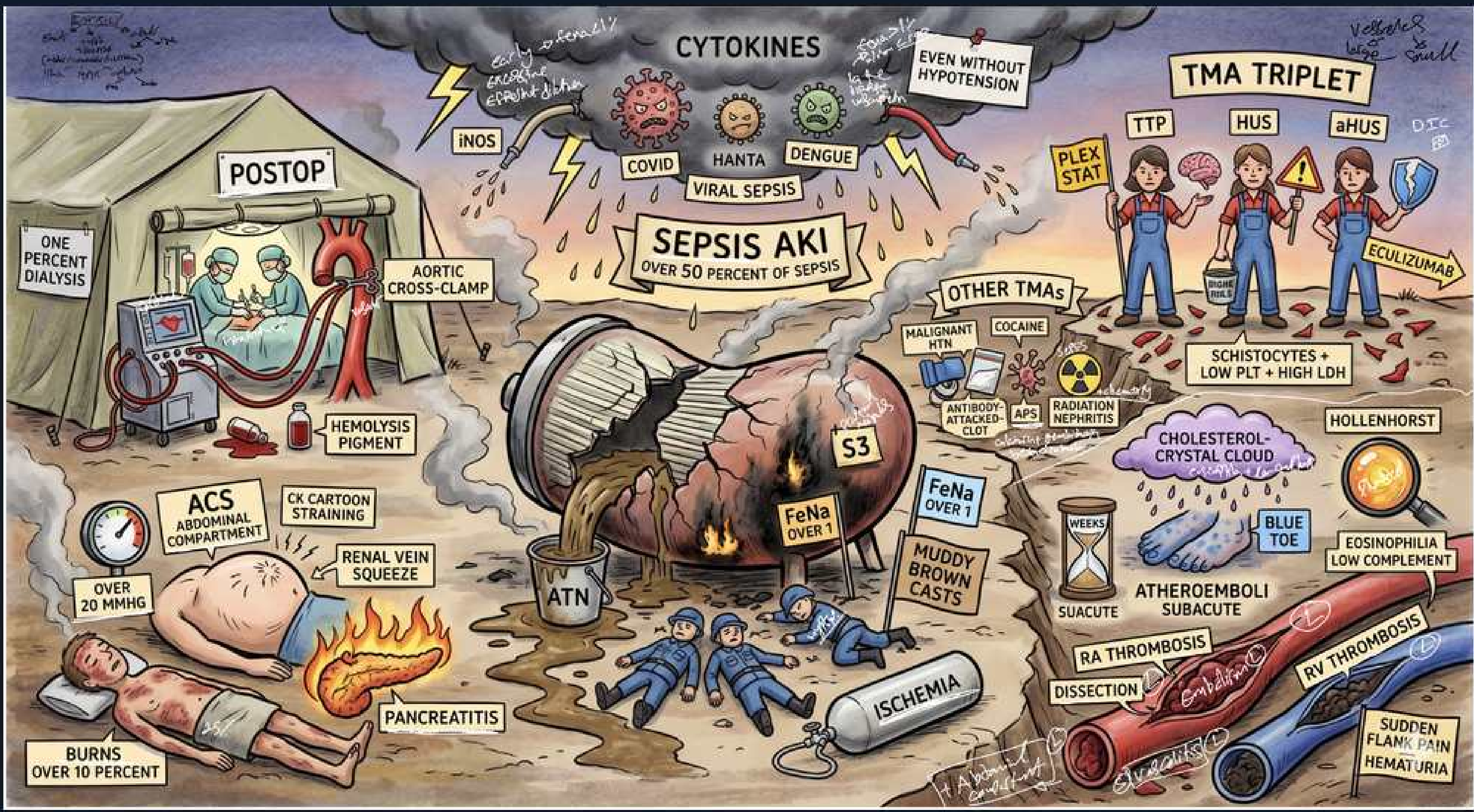
AKI — Intrinsic 1: ATN



1. ATN

WHAT YOU SEE

Cracked tubule barrel, ATN sign, ischemia figures

WHAT IT ENCODES

Acute tubular necrosis (ATN) is the most common cause of INTRINSIC AKI, from ischemic or toxic death of tubular epithelial cells (most severe in the proximal tubule and the medullary thick ascending limb, the most metabolically active/hypoxia-prone segments). Sloughed cells form casts that obstruct tubules and back-leak filtrate. Course has three phases: initiation, maintenance (oliguric, lasting 1–3 weeks), and recovery (polyuric).

BOARD TRAP

ATN is confused with prerenal azotemia because both follow hypoperfusion — the discriminator is reversibility and labs: prerenal corrects rapidly with fluids and has FENa <1%, BUN:Cr >20:1, bland urine; established ATN does NOT correct quickly and shows FENa >2% with muddy-brown casts.

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2. Muddy brown casts

WHAT YOU SEE

MUDDY BROWN CASTS sign

WHAT IT ENCODES

The urinary hallmark of ATN is muddy-brown granular 'dirty' casts and renal tubular epithelial cell casts, formed from sloughed necrotic tubular cells. Their presence strongly supports ATN over prerenal AKI (which has a bland sediment).

BOARD TRAP

A student picks WBC casts as diagnostic of ATN — WRONG. WBC casts indicate pyelonephritis or AIN; RBC casts indicate glomerulonephritis. ATN = MUDDY-BROWN GRANULAR casts. Mixing up the cast types is a high-yield trap.

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3. FENa >1-2%

WHAT YOU SEE

FENa OVER 1 sign

WHAT IT ENCODES

In ATN, FENa is >2% (and urine osmolality <350, urine Na >40) because damaged tubules lose the ability to reabsorb sodium and concentrate urine. FENa = (UNa × PCr)/(PNa × UCr) × 100. Prerenal AKI by contrast has FENa <1% as intact tubules avidly retain sodium.

BOARD TRAP

Two big traps: (1) thinking established ATN has FENa <1% — that is prerenal; (2) trusting FENa in a patient on DIURETICS, where FENa is falsely elevated — use FEUrea instead (<35% suggests prerenal). FENa can also be <1% in early ATN, contrast nephropathy, rhabdomyolysis, and hepatorenal syndrome despite tubular injury.

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4. Ischemia

WHAT YOU SEE

ISCHEMIA sign under fallen figures

WHAT IT ENCODES

Ischemic ATN results when prolonged prerenal hypoperfusion (sepsis, hemorrhage, hypotension, shock, cardiac arrest) is severe or sustained enough to cause actual tubular cell death — the continuum from prerenal azotemia to ATN. Restoring perfusion does not immediately reverse it; recovery takes weeks.

BOARD TRAP

Assuming a hypotensive patient with rising creatinine just needs fluids (prerenal) — if hypoperfusion was prolonged, it has already become ATN, which will NOT respond promptly to volume. The discriminator is duration of insult plus urine findings.

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5. Sepsis AKI

WHAT YOU SEE

SEPSIS AKI banner, 'over 50% of sepsis', cytokines

WHAT IT ENCODES

Sepsis is the leading cause of ATN in hospitalized patients, contributing to AKI in over 50% of severe sepsis. Injury occurs through inflammatory cytokines, microvascular dysfunction, and tubular stress — and importantly can happen even WITHOUT documented systemic hypotension.

BOARD TRAP

Requiring documented hypotension before attributing AKI to sepsis is WRONG — septic AKI develops via cytokine-mediated and microvascular injury even with normal blood pressure. So a normotensive septic patient with rising creatinine still has sepsis-associated AKI.

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6. Aortic cross-clamp

WHAT YOU SEE

POSTOP, aortic cross-clamp surgery

WHAT IT ENCODES

Postoperative ischemic ATN classically follows cardiac and aortic surgery, where aortic cross-clamping interrupts renal arterial perfusion. Cardiopulmonary bypass, large fluid shifts, hemolysis, and atheroemboli compound the risk.

BOARD TRAP

Postop AKI after vascular/aortic surgery may be ATN from ischemia OR atheroembolic disease from plaque disruption — distinguish by timing and findings (atheroemboli give blue toes, livedo, eosinophilia, low complement; pure ischemic ATN does not).

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7. Rhabdo / myoglobin

WHAT YOU SEE

Hemolysis pigment, CK cartoon straining

WHAT IT ENCODES

Pigment nephropathy: in rhabdomyolysis, released myoglobin is directly toxic to tubules, causes tubular obstruction, and triggers vasoconstriction → ATN. Clues: CK often >5,000–10,000 (markedly elevated), dipstick POSITIVE for blood but FEW RBCs on microscopy (heme pigment, not cells), plus hyperkalemia, hyperphosphatemia, hypocalcemia, and elevated uric acid. Treat with aggressive IV isotonic fluids.

BOARD TRAP

A urine dipstick positive for blood with no RBCs on microscopy is misread as a bad sample — WRONG; it signals myoglobinuria (or hemoglobinuria). Also, FENa can be <1% in pigment nephropathy despite true tubular injury, so don't use it to call this prerenal.

8. Crush / burns

WHAT YOU SEE

BURNS over 10 percent, pancreatitis, crush figure

WHAT IT ENCODES

Burns (especially >10–20% BSA), crush injury, and pancreatitis cause ATN through combined mechanisms: massive third-space/volume loss (prerenal → ischemic ATN) plus pigment release (myoglobin in crush) and a systemic inflammatory response.

BOARD TRAP

Underestimating fluid needs in burns/crush leads to under-resuscitation and ATN — but over-resuscitation causes abdominal compartment syndrome, itself a cause of AKI. The discriminator is careful volume titration, not a fixed rule.

9. Abd compartment

WHAT YOU SEE

ACS abdominal compartment, OVER 20 MMHG

WHAT IT ENCODES

Abdominal compartment syndrome (sustained intra-abdominal pressure >20 mmHg with new organ dysfunction) compresses the renal veins and parenchyma, raising renal venous pressure and dropping renal perfusion pressure → oliguric AKI. Measured via bladder pressure; definitive treatment is decompression.

BOARD TRAP

Oliguria in a tense, distended postoperative/trauma abdomen is treated with more fluids — WRONG, that worsens compartment pressure. The discriminator is measuring bladder (intra-abdominal) pressure; treatment is decompression, not volume.

10. TMA triplet

WHAT YOU SEE

TMA TRIPLET: TTP / HUS / aHUS three figures

WHAT IT ENCODES

Thrombotic microangiopathies cause intrinsic AKI via platelet-fibrin microthrombi in glomerular capillaries: TTP (ADAMTS13 deficiency, more neuro/CNS-predominant), typical HUS (Shiga-toxin E. coli O157:H7, more renal-predominant, often after bloody diarrhea in children), and atypical HUS (complement dysregulation). TTP is treated with plasma exchange; aHUS with eculizumab.

BOARD TRAP

Treating suspected TTP/HUS with platelet transfusion is the classic wrong answer — it can fuel further thrombosis. Also, do NOT give antibiotics for Shiga-toxin HUS, as they may increase toxin release. The discriminator: plasma exchange for TTP, supportive care for typical HUS, eculizumab for aHUS.

11. Schistocytes

WHAT YOU SEE

Schistocytes + low PLT + high LDH

WHAT IT ENCODES

TMA labs show microangiopathic hemolytic anemia (MAHA): schistocytes on smear, thrombocytopenia, elevated LDH and indirect bilirubin, low haptoglobin, and a NEGATIVE direct Coombs (mechanical, not antibody-mediated, hemolysis).

BOARD TRAP

A Coombs-POSITIVE hemolytic anemia points to autoimmune hemolysis, NOT a TMA — the negative Coombs plus schistocytes is the discriminator. Don't confuse TMA with DIC, which also has schistocytes but ADDS abnormal PT/PTT and low fibrinogen (TMA coags are normal).

12. Atheroemboli

WHAT YOU SEE

Cholesterol crystal cloud, BLUE TOE, Hollenhorst plaque

WHAT IT ENCODES

Atheroembolic (cholesterol crystal) AKI occurs DAYS TO WEEKS after an arterial procedure (cardiac cath, angiography, vascular surgery) or anticoagulation, when plaque cholesterol crystals shower distally. Classic triad: subacute/stepwise AKI, livedo reticularis with blue toes (with palpable pulses), and eosinophilia. Other clues: low complement, Hollenhorst plaques on funduscopy.

BOARD TRAP

This is the great mimic of contrast nephropathy — the discriminator is TIMING and signs: contrast nephropathy peaks at 3–5 days and resolves in 7–10, whereas atheroembolic AKI starts days-to-weeks later, progresses stepwise, and adds blue toes/livedo, eosinophilia, and low complement.

13. Contrast / drugs

WHAT YOU SEE

Other TMAs: cocaine, antibody MTX, radiation nephritis

WHAT IT ENCODES

Nephrotoxic ATN from direct tubular toxins: iodinated contrast (contrast-induced nephropathy via vasoconstriction + tubular toxicity, peaks day 3–5), aminoglycosides, amphotericin B, cisplatin, and others; drug-induced TMA (e.g., calcineurin inhibitors, gemcitabine, quinine, cocaine-adulterants) and radiation nephritis are additional mechanisms.

BOARD TRAP

Contrast nephropathy is overdiagnosed — a transient creatinine bump after contrast may instead be atheroembolic disease (later, stepwise, blue toes/eosinophilia). The discriminator is the time course and associated findings, not the mere fact contrast was given.

14. PV thrombosis

WHAT YOU SEE

RA thrombosis / PV thrombosis vessels, dissection

WHAT IT ENCODES

Renovascular causes of acute intrinsic AKI: bilateral renal artery thromboembolism/dissection and renal vein thrombosis. Renal vein thrombosis is classically associated with NEPHROTIC SYNDROME (especially membranous nephropathy) due to urinary antithrombin loss and hypercoagulability.

BOARD TRAP

Acute flank pain + hematuria + AKI is reflexively called a kidney stone — but in a nephrotic patient it may be renal vein thrombosis. The discriminator is the hypercoagulable/nephrotic context plus imaging (CT/Doppler), not assuming nephrolithiasis.

15. Viral sepsis triggers

WHAT YOU SEE

COVID, HANTA, DENGUE viral icons feeding the sepsis storm cloud

WHAT IT ENCODES

Viral infections are increasingly recognized causes of severe AKI: COVID-19 (cytokine storm, direct tubular injury, often with proteinuria/collapsing FSGS in APOL1-risk patients), hantavirus (hemorrhagic fever with renal syndrome — fever, AKI, thrombocytopenia), and dengue. These drive AKI via the same sepsis/cytokine pathway as bacterial sepsis.

BOARD TRAP

Attributing AKI in a febrile traveler or pandemic patient solely to dehydration misses viral cytokine-mediated ATN — the discriminator is the viral syndrome (rash, thrombocytopenia, exposure history) plus AKI that outlasts volume repletion.

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16. One percent dialysis

WHAT YOU SEE

ONE PERCENT DIALYSIS sign at the postop tent

WHAT IT ENCODES

Most ATN is managed supportively with the expectation of tubular regeneration over 1–3 weeks; only a minority (often cited around 1% in some series, more in critically ill cohorts) of ATN/postop AKI patients ultimately require dialysis. There is no drug that reverses established ATN — care is supportive while waiting for recovery.

BOARD TRAP

Reaching for 'renal-protective' agents (low-dose dopamine, mannitol, loop diuretics, fenoldopam) to prevent or treat ATN is the wrong answer — none improve recovery or mortality; they may convert oliguric to non-oliguric ATN without changing outcome.

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17. Sudden flank pain + hematuria

WHAT YOU SEE

SUDDEN FLANK PAIN + HEMATURIA banner at the vascular corner

WHAT IT ENCODES

Acute renal infarction (from arterial thromboembolism — e.g., atrial fibrillation, endocarditis — or dissection) presents with abrupt flank/abdominal pain, hematuria, and a markedly elevated LDH out of proportion to other findings. Diagnosis is by contrast CT angiography showing a wedge-shaped perfusion defect.

BOARD TRAP

This presentation is misdiagnosed as nephrolithiasis or pyelonephritis — the discriminators are a very high LDH, an embolic source (AFib), and a wedge-shaped infarct on CT angiography rather than a stone or infection.
